Trenitalia
ETR1000 VZI-6

Diagnostics
VZI-6

Revision	Author	Date	Revision Description
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00.2	G. Iusto	22/05/2023	First Release after Internal Review
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1 Introduction

This document is a function specification for the control of the *Diagnostic* system for VZI-6 for customer Trenitalia.

1.1 Train Configuration

Each VZI-6 train consists of eight cars in a fixed formation including a driver's cab at each end:

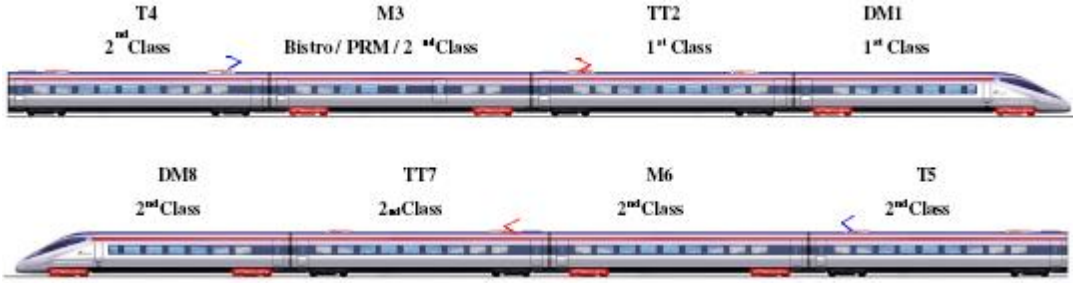


Figure 1: Train Architecture

- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

1.2 Definitions and abbreviations

CD	Circuit Diagram
CCU	Central Computing Unit
MDCU	Central Computing Unit – Regular and Safe Diagnostic functions
CCU-R	Central Computing Unit – Regular Operational Functions
CCU-S	Central Computing Unit – Safe Operational Functions
ECN	Ethernet Consist Network
EVENT	Diagnostics that will be stored in the Event database within TCMS system.
GUI	Graphical User Interface
HW HMI	Hardware Human Machine Interface, e.g. buttons, switches and indication lamps.
TOD	Train operator Display / Interface Display Unit
ICD	Interface Control Document
IPT	Internet Protocol – Train
MDS	Monitoring and Diagnostic Data System
ODBS	Onboard Diagnostic Database System
PCL	Process Client
PTU	Portable Test Unit
TCL	Tool Client
TCMS	Train Control and Management System
MDS	Train Diagnostic System
MDS-DM	Data and Monitoring of the Train Diagnostic System
MDS-DR	Data Recorder of the Train Diagnostic System
TOD	Train Operator Display
TSSP	Train Subscription Service Provider
VRS	Version Reporting System
WBS	Work Breakdown Structure

2 System Overview

The purpose of the TCMS MDS (Train Diagnostic System) is to ascertain and provide the current status of the vehicle by gathering status data among the vehicle systems. Diagnostic messages are generated from both TCMS equipment as well as sub-suppliers equipment and stored by the diagnostic system. In case an event occurs which causes an operational impact to the vehicle it will be presented to the user combined with user advice.

2.1 System Configuration

The on-board part of the train diagnostic system, MDS, is a sub-system of the TCMS. The TCMS MDS is located in the HMD-06 (pos.1) and in the HMD-02 (pos.2) cards of the CDU-SIL Controller as shown in the following Figure.

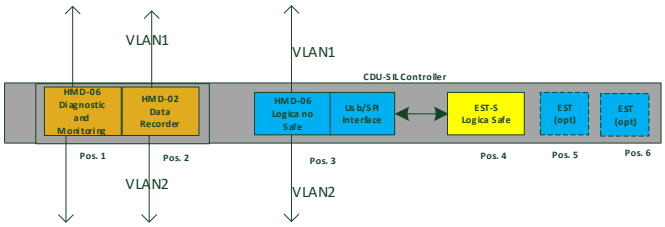


Figure 2: CDU-SIL Controller Architecture

- The HMD-06 (pos.1) card has in charge the Diagnostic and Monitoring functions (MDS-DM) of the Train Diagnostic System.
  - It collects and acts upon relevant information about events, faults and vehicle status from the connected systems;
  - It stores fault data and environment data in the Diagnostic Data Base in a Non-Volatile support;
  - In addition, it drives the HMI devices (TOD units) to present information to the Driver, to the Train Crew operators and to the on board Maintenance people.

- The HMD-02 (pos2) card has in charge the Data Recorder function (TRS-DR) of the Train Diagnostic System
  - It continuously stores in a Non-Volatile support all the train variables deemed of interest for off-board analyses

Stored Data can be uploaded to an off-board maintenance and service system for long term storage and additional off line analysis.

The following figure schematically shows the architecture of the train diagnostic system and the data flows on the communication bus:

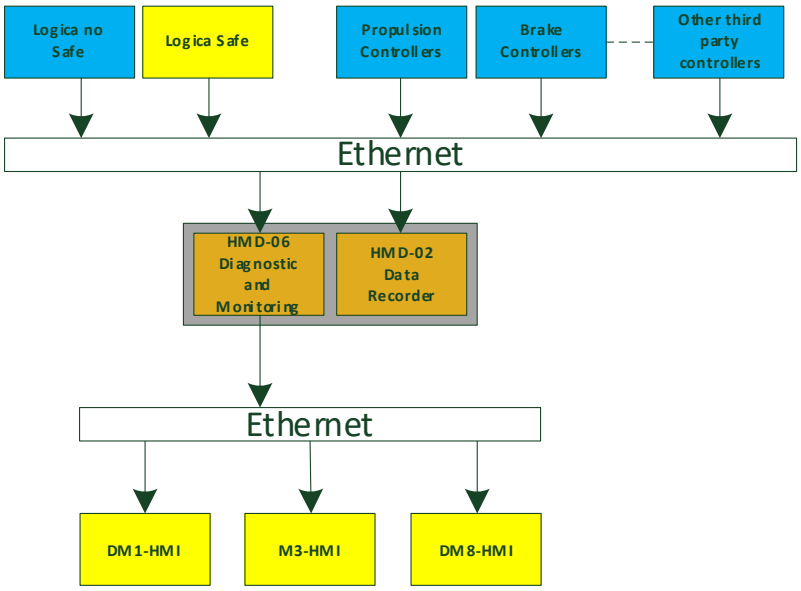


Figure: 3 Train Diagnostic System (MDS)

For failure mode and redundancy concept refer to SFRS TCMS Operational.

3 Functional requirements

3.1 Diagnostics

3.1.1 Function General requirements

3.1.1.1 *Function Design*

The MDS collects all the diagnostic data generated by the other control units and essentially performs the following functions:
1) It stores diagnostic data in non-volatile memory
2) Defines relevant diagnostic messages to present to the driver based on collected data and priorities
3) It defines the Train Status data based on the collected information
4) It drives the human machine interface equipment (TOD)
5) It continuously stores the train data

Requirements

ID	2F_10.01.Zefiro-Europe.TRS.66
Description	Events shall cover all systems listed in below including TCMS itself: - Air Supply System - ATP System - Auxiliary Electric System - Battery System - Brake System - CCTV System

	- Driver's Safety Device - Exterior Doors - Fire Detection System - Heating, Ventilation and Air Conditioning System - High Voltage System - Infotainment - Interior Doors - Interior Lighting - Passenger Information System - Propulsion System - Toilets - Bogie Monitoring and other systems.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
[2F_10.01.Zefiro-Europe.CONCEPT-NEW-1]	Following the failure events generated by the TCMS system and by third-party control units, the MDS must identify the primary failure event, to be presented to the driver, and filter the secondary events resulting from the primary event. Rationale: The primary event is the event identifying the anomaly actually present, the secondary events are generated by the control unit logics as a consequence of the primary event and are not significant for the purposes of the indications to be given to the driver for continuing the mission. In any case, secondary events are normally stored in non-volatile memory for analysis ashore.	Derived Requirement			N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.120	The TCMS system shall implement the event reporting for	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

	the events defined in the different systems connected to TCMS and defined in the system SFRS				
2F_10.01.Zefiro-Europe.CONCEPT.360	The TCMS system shall implement the event reporting for the events defined in the different systems connected to TCMS and defined in the system SFRSes Note: Events front all coupled train-sets shall be visible to the driver.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_10.01.Zefiro-Europe.TRS.96
Description	Gathered event data and condition data shall be stored in a database located in battery backed up memory or with equivalent security so it can be used for operational, maintenance or vehicle performance measurements purposes.
Safety level	N/A

ID	2F_10.01.Zefiro-Europe.TRS.129
Description	The diagnostic system shall be redundant.
Safety level	N/A

MDCU / ODBS Firmware

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.146	The ODBS shall be configured to be stored in non volatile memory.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_10.01.Zefiro-Europe.CONCEPT.322	Redundant ODBS shall be supported to maintain functionality under any single point of failure.	Derived Requirement	4S_09.03.06.-.TCMS Hardware		N/A

ID	2F_10.01.Zefiro-Europe.TRS.100
Description	
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.155		Derived Requirement	4S_09.03.06.-.TCMS Hardware		N/A

3.1.2 Function Event handling

3.1.2.1 *Function Design*

Relating to the events a set of attributes needs to be defined that controls how events can be presented on the TOD as well as the descriptive text elements associated with the events.

3.1.2.2 *Requirements*

ID	2F_10.01.Zefiro-Europe.TRS.103
Description	Each event shall have the following attributes: - Code/identity - Category - Start time - End time - - Texts
Safety level	N/A

ID	2F_10.01.Zefiro-Europe.TRS.92
Description	The events shall be classified based on severity and character.
Safety level	N/A

ID	2F_10.01.Zefiro-Europe.TRS.98
Description	MDS shall support multiple language (, Italian, English) event texts.
Safety level	N/A

ID	2F_10.01.Zefiro-Europe.TRS.99
Description	Italian shall be the default language. It shall be possible to switch between languages in run-time.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL

Event priority

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.163	For each event a priority shall be set Note: TCMS team responsible for first proposal.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_10.01.Zefiro-Europe.CONCEPT.165	Each event shall be classified based on severity, character and customer expectations.	Derived Requirement	5X_DOC.-.Documentation and Training		N/A

Event texts

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.140	TCMS MDS shall be configured for four different languages (, Italian, English).	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_10.01.Zefiro-Europe.CONCEPT.150	4S_09.03.05.-.TCMS Software		N/A		
2F_10.01.Zefiro-Europe.CONCEPT.152	For each event a short event description, extended event description (page 1) and a maintainer event description (page 2) shall be defined. Note: Supplier team responsible for first proposal of the short event description.	Derived Requirement	4S_05.-.Propulsion		N/A
2F_10.01.Zefiro-Europe.CONCEPT.153	The short event description, extended event description (page 1) and a	Derived Requirement	5X_DOC.-.Documentation and Training		N/A

	maintainer event description (page 2) shall be defined in, Italian and English according to agreement with the customer.				

Event code / identity

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.295	For each diagnostic event, the MDS shall define an event code that is unique within each consist diagnostic database.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_10.01.Zefiro-Europe.CONCEPT.297	Subsystem groups shall be structured to fit the needs for the TOD active event overview menu.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

General event attributes

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.299	Events in the MDS ODBS shall be stored with start time, end time	Derived Requirement	4S_09.03.06.-.TCMS Hardware		N/A

ID	2F_10.01.Zefiro-Europe.TRS.101
Description	The MDS module shall provide the global environmental signals that are generic for all events.
Safety level	N/A

ID	2F_10.01.Zefiro-Europe.TRS.68
Description	Environmental data with pre- and post-history shall be attatched to each event.
Safety level	N/A

Environment data

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.157	A set of environment data	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

	shall be defined and used for all generated events,.				
2F_10.01.Zefiro-Europe.CONCEPT.170	Number of pre- and post samples of environmental data shall be configured per PCL.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_10.01.Zefiro-Europe.TRS.97
Description	The location and origin of collected data shall be visible. It shall be possible to determine car and unit.
Safety level	N/A

Event location

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.148	The PCL Location ID shall be set to values depending on car for the reported TCMS events. Car Location ID DM1 1 TT2 2 M3 3 T4 4 T5 5 M6 6 TT7 7 DM8 8	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

3.1.3 Function Event categorization

3.1.3.1 *Function Design*

Events are assigned to different categories. These categories are also used to control the behaviour of how events are presented on the TOD (as described in SFRS Visualization).

3.1.3.2 *Requirements*

ID	2F_10.01.Zefiro-Europe.TRS.70
Description	Events shall use the following categories listed in priority order: 0. Serious falt events Information to the driver about the serious fault that has occurred. Automatically displayed on TOD when it has occured. 1. Advisory

	Information to the driver that he has done an incorrect maneuver. No acknowledge for this event. Automatically displayed on TOD when it has occured. 2. Alarm Major event, i.e. stopping events that require the relevant user's full attention. Automatically displayed on TOD when it has occured. 3. Warning Warnings or service effecting event, i.e. train performance reduction event. Automatically displayed on TOD when it has occured. 4. Maintenance Minor event intended for maintenance personnel. No service effecting event. Available in event lists on TOD for maintainers.. 5. Other Informative Event, Non effecting event. Available in event history list on TOD.
Safety level	N/A

ID	2F_10.01.Zefiro-Europe.TRS.105
Description	Events with higher priority shall take precedence over events with lower priority, even if a lower priority event has already been presented on the TOD.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.224	Events shall be categorized into following event groups: Serious fault events, Advisory, Alarm, Warning, Maintenance, Other, with the priorities set according to (from highest priority to lowest): Serious fault events Advisory Alarm Warning Maintenance Other	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_10.01.Zefiro-Europe.CONCEPT.223	For test purposes one event of each type shall be implemented in a way (connected to a software parameter) that makes it easy to set the events in a test set-up.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_10.01.Zefiro-Europe.TRS.131
Description	Serious fault events shall be presented on the TOD and they shall be acknowledged by pressing a button on the TOD. When the event has been acknowledged it shall be removed from the screen.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.341	Serious fault events shall be presented on the TOD and shall be acknowledged using a button on the TOD.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A

ID	2F_10.01.Zefiro-Europe.TRS.106
Description	Advisory event shall be displayed on the TOD. As soon as the driver corrects the fault, the Advisory event shall disappear automatically from the screen.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.331	Advisory event shall be displayed on the TOD as long as the incorrect handling is active. It shall disappear automatically when the incorrect action is corrected.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A

ID	2F_10.01.Zefiro-Europe.TRS.73
Description	It shall be possible for the driver, crew and maintenance personnel to see all active events on every TOD in the train (relevant for user category).
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.330	An active events list shall be provided for each user category, ie. driver or maintainer.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A

ID	2F_10.01.Zefiro-Europe.TRS.130
Description	Alarm events shall be presented on the TOD and they shall be acknowledged by pressing a button on the TOD before it is removed from the screen.

Safety level	N/A

Nr	**Description**	**Applicability Details**	**Derived to**	**User Interface**	**SIL**
2F_10.01.Zefiro-Europe.CONCEPT.329	Alarm events shall always require acknowledgement on the TOD, the alarm message shall be removed from the screen only after acknowledgement.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A

ID	2F_10.01.Zefiro-Europe.TRS.110
Description	Warning events shall be presented on the TOD and they shall be acknowledged by pressing a button on the TOD before it is removed from the screen.
Safety level	SIL 0

Nr	**Description**	**Applicability Details**	**Derived to**	**User Interface**	**SIL**
2F_10.01.Zefiro-Europe.CONCEPT.327	Warning events shall be presented on the TOD and they shall be acknowledged by pressing a button on the TOD before it is removed from the screen.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

ID	2F_10.01.Zefiro-Europe.TRS.107
Description	Maintenance personal shall be able to acknowledge all events both indivTODally and as a group except for serious fault events that shall be acknowledged indivTODally.
Safety level	N/A

Nr	**Description**	**Applicability Details**	**Derived to**	**User Interface**	**SIL**
2F_10.01.Zefiro-Europe.CONCEPT.328	If the TOD is activated by a maintainer (maintanance mode), the personal shall be able to acknowledge all events both indivTODally and as a group except for Serious fault events that shall be	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

	acknowledged indivTODally.				
2F_10.01.Zefiro-Europe.CONCEPT.348	The driver shall be able to acknowledge all events once, when CCUO (no-safe Active Vehicle Logic function) allowed to do that.	Derived Requirement			
2F_10.01.Zefiro-Europe.CONCEPT.361	The driver shall be able to acknowledge all events once, when TCMS system allowed to do that.	Derived Requirement	4S_09.03.05.-.TCMS Software		

3.1.4 Buzzer interface

3.1.4.1 *Function Design*

Critical events will sound a buzzer in the cab to notify the driver.

3.1.4.2 *Requirements*

ID	2F_10.01.Zefiro-Europe.TRS.76
Description	New Alarm event shall trigger the buzzer in the active cab.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.228	Each new Alarm event shall give a trigger to the buzzer in the active cab. Triggers shall not be queued.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_10.01.Zefiro-Europe.CONCEPT.334	The Buzzer in the cab shall sound for 5s and for 1s every 5s for maximm 15s or until it is acknowledged by the user by selecting the acknowledge button on the event field on the TOD.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

3.1.5 Function Condition data handling

3.1.5.1 *Function Design*

Condition data is used to store non-volatile parameters and counters as defined in the different system SFRS. The collection in this section only defines the base functionality to use for such condition data.

3.1.5.2 *Requirements*

ID	2F_10.01.Zefiro-Europe.TRS.78
Description	Condition data of type parameters and counters shall be supported.
Safety level	N/A

ID	2F_10.01.Zefiro-Europe.TRS.94
Description	The diagnostic system shall also store other data (called condition data) about the various systems such as operation time and number of operations.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.142	The TCMS system shall implement condition data (parameters and counters) as defined in the different systems connected to TCMS and defined in the system SFRSes.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_10.01.Zefiro-Europe.CONCEPT.304	Condition data of type parameters shall have a default value to be used as start value.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_10.01.Zefiro-Europe.CONCEPT.305	Condition data of type counters shall have a counter limit to be used to compare the counter value.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

3.1.6 Diagnostics data set at train level

3.1.6.1 *Function Design*

3.1.6.2 *Requirements*

Nr	Description	Applicability Details	Derived to	User Interface	SIL

2F_10.01.Zefiro-Europe.CONCEPT.345	Before every TCMS shutdown an event shall be set, in order to indicate that the train has been requested to power off	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_10.01.Zefiro-Europe.CONCEPT.346	When a user log in as a Maintainer in the TOD an event shall be set, in order to indicate that the train is under maintenance	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_10.01.Zefiro-Europe.CONCEPT.347	When a user log out from Maintainer in the TOD an event shall be set, in order to indicate that the train is no more under maintenance	Derived Requirement	4S_09.03.05.-.TCMS Software		

3.2 Function Software version handling

3.2.1 Function Design

The version reporting system consists of a server that stores version information reported by the different TCMS devices. The TCMS application software, by use of information provided by connected systems, reports also version information for those. The TOD and MCG additionally include requester to get version information from the server.

3.2.2 Requirements

ID	2F_10.01.Zefiro-Europe.TRS.80
Description	The TCMS system shall include a version reporting system used for application software version (for presentation on the TOD).
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.244	The TCMS system shall include a VRS server.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_10.01.Zefiro-Europe.CONCEPT.245	The TCMS controllers shall include VRS client and software/hardware reporters to send version/revision data to the VRS server	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

	for the indivTODal controller.				
2F_10.01.Zefiro-Europe.CONCEPT.362	The TCMS system shall include VRS client and software/hardware reporters to send version/revision data to the VRS server for the indivTODal controller.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_10.01.Zefiro-Europe.CONCEPT.291	The TCMS system shall implement generic VRS reporters for RIO-units.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_10.01.Zefiro-Europe.CONCEPT.306	The TCMS system shall implement generic VRS reporters for systems connected to TCMS, to report software versions, serial number and hardware revisions. Note: Information defined in the different system Interface Control Documents.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_10.01.Zefiro-Europe.CONCEPT.308	The TOD shall include a VRS requester to access the stored VRS data.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_10.01.Zefiro-Europe.TRS.117
Description	The stored software versions shall be possible to upload to a connected service LapTop and Remote (via MCG).
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.246	It shall be possible to download the stored VRS data from a service LapTop connected to ECN via a	Derived Requirement	4S_09.03.06.-.TCMS Hardware		N/A

	service access point.				
2F_10.01.Zefiro-Europe.CONCEPT.247	The MCG shall provide a service to download VRS data from the train.	Derived Requirement	4S_09.03.06.-.TCMS Hardware		N/A

3.3 Maintenance

3.3.1 Service Access

3.3.1.1 *Function Design*

Service access is performed from a service laptop PC connected to a service access point on an IP ring switch.

3.3.1.2 *Requirements*

ID	2F_10.01.Zefiro-Europe.TRS.116
Description	From Each cab IP service port there is possible to download diagnostic data, to upload software, check communication, configuration, software version numbers, monitor and temporary changing parameters and initiate function tests.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.259	In each cab there shall be an easy accessible service port connected to an IP switch.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_10.01.Zefiro-Europe.CONCEPT.332	From Each IP service port it shall be possible to upload diagnostic data, download software, check communication, configuration, software version numbers, monitor and temporary changing parameters and initiate function tests.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_10.01.Zefiro-Europe.TRS.119
Description	It shall be possible to view signals and parameters from a connected service LapTop.
Safety level	N/A

ID	2F_10.01.Zefiro-Europe.TRS.120
Description	It shall be possible to temporarly change parameters in the TCMS software from a connected service LapTop.
Safety level	N/A

ID	2F_10.01.Zefiro-Europe.TRS.123
Description	It shall be possible to analyze signals status over time in graphical format.
Safety level	N/A

Nr	**Description**	**Applicability Details**	**Derived to**	**User Interface**	**SIL**
2F_10.01.Zefiro-Europe.CONCEPT.285	It shall be possible to view signals and parameters in all devices applications, using a PTU on a service LapTop connected to ECN via a service access point.	Derived Requirement	4S_09.03.06.-.TCMS Hardware		N/A
2F_10.01.Zefiro-Europe.CONCEPT.316	It shall be possible to temporary change parameters, using the PTU on a service LapTop connected to ECN via a service access point.	Derived Requirement	4S_09.03.06.-.TCMS Hardware		N/A
2F_10.01.Zefiro-Europe.CONCEPT.317	The PTU shall support logmonitor, fast log and trace window to enable analysis of signals over time in graphical format.	Derived Requirement	4S_09.03.06.-.TCMS Hardware		N/A
2F_10.01.Zefiro-Europe.CONCEPT.318	The PTU shall support GUI in the following languages Italian and English.	Derived Requirement	4S_09.03.06.-.TCMS Hardware		N/A

3.3.2 Software Download

3.3.2.1 *Function Design*

Download of software to the TCMS system and to verify that correct files are downloaded.

3.3.2.2 *Requirements*

ID	2F_10.01.Zefiro-Europe.TRS.61
Description	It shall be possible to download application software and other information to the TCMS units (including the DCU/A, /F, /L, /M and LIM) from a connected service LapTop.
Safety level	SIL 0

ID	2F_10.01.Zefiro-Europe.TRS.122
Description	Using a PC it shall be possible the uploading of a new Software version and the check of uploaded applications.
Safety level	N/A

Nr	**Description**	**Applicability Details**	**Derived to**	**User Interface**	**SIL**
2F_10.01.Zefiro-Europe.CONCEPT.269	It shall be possible to download software (operating system, applications, configuration files) to the TCMS controllers connected to ECN, using a service LapTop connected to ECN via a service access point.	Derived Requirement	4S_09.03.06.-.TCMS Hardware		N/A
2F_10.01.Zefiro-Europe.CONCEPT.320	The propulsion system shall include needed functionality to support software download over ECN to all included devices. Note: This also includes the LIM	Derived Requirement	4S_05.-.Propulsion		N/A
2F_10.01.Zefiro-Europe.CONCEPT.309	It shall be possible verify that correct versions have been dowloaded to the controllers.	Derived Requirement	4S_09.03.06.-.TCMS Hardware		N/A
2F_10.01.Zefiro-Europe.CONCEPT.310	It shall be possible to check what	Derived Requirement	4S_09.03.06.-.TCMS Hardware		N/A

	versions are downloaded to the controllers without performing a new download operation.				
2F_10.01.Zefiro-Europe.CONCEPT.311	The Software download facility shall support GUI in the following languages Italian and English.	Derived Requirement	4S_09.03.06.-.TCMS Hardware		N/A

3.3.3 Function Upload and analysis

3.3.3.1 *Function Design*

Upload of diagnostic data and performing analysis of the uploaded data.

3.3.3.2 *Requirements*

ID	2F_10.01.Zefiro-Europe.TRS.95
Description	It shall be possible to perform remote upload of the stored diagnostic data to an off-board system.
Safety level	N/A

ID	2F_10.01.Zefiro-Europe.TRS.46
Description	The event information stored in the train diagnostic database shall be possible to upload to an off-board maintenance and service system for long term storage and additional off line analysis.
Safety level	N/A

ID	2F_10.01.Zefiro-Europe.TRS.112
Description	The diagnostic system shall interface the Train Telediagnostica System for remote upload of stored event data.
Safety level	N/A

Nr	**Description**	**Applicability Details**	**Derived to**	**User Interface**	**SIL**
2F_10.01.Zefiro-Europe.CONCEPT.312	The MDS upload tool shall support GUI in the following languages: Italian and English.	Derived Requirement	4S_09.03.06.-.TCMS Hardware		N/A
2F_10.01.Zefiro-Europe.CONCEPT.333	Upload of diagnositc data to the Telediagnostica	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

	application shall be supported.				

ID	2F_10.01.Zefiro-Europe.TRS.121
Description	r The MDS shall support the analysis of uploaded events on a sevice LapTop.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.272	It shall be possible to upload MDS data, using an application on a service LapTop connected to ECN via a service access point.	Derived Requirement	4S_09.03.06.-.TCMS Hardware		N/A
2F_10.01.Zefiro-Europe.CONCEPT.313	It shall be possible to analyse the uploaded data on the service LapTop	Derived Requirement	4S_09.03.06.-.TCMS Hardware		N/A
2F_10.01.Zefiro-Europe.CONCEPT.314	The LapTop application Program for analisys shall support GUI in the following languages: Italian and English.	Derived Requirement	4S_09.03.06.-.TCMS Hardware		N/A

3.3.4 Function Subscription of signals

3.3.4.1 *Function Design*

Base functions for the TOD as well as train to wayside communication for accessing signals from the TCMS system.

3.3.4.2 *Requirements*

ID	2F_10.01.Zefiro-Europe.TRS.115
Description	It shall be possible to subscribe to sub-system signals on TRDP, e.g. from the MDCU.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.273	TOD shall use TRDP communication to subscribe to sub-system signals.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_10.01.Zefiro-Europe.TRS.114
Description	
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.274	It shall be possible to subscribe to onboard TRDP signals from a ground based system (via the Train Telediagnostica System).	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

4 Signal Interface

4.1 I/O Interface

4.1.1 Supplier system

Inputs:

Nr	Functional name	Type	Linked to requirement

Outputs:

Nr	Functional name	Type	Linked to requirement

4.2 Bus Interface

Inputs:

Nr	Functional name	Type	Linked to requirement
	The Diagnostic System shall read the TRDP datasets specified in the ICD documents defined during the development		

Outputs

Nr	Functional name	Type	Linked to requirement
	The Diagnostic System shall write the TRDP datasets specified in the ICD documents defined during the development		